Department of Shipping

Inland Ship Safety Administration

141-143, Motijheel C/A (8th Floor)

Dhaka-1000.

Application form

for the Approval of Design / Plan

for Construction / Reconstruction

of Inland Ship.

MOHAMMAD HOSAIN SUMON

Ref. No. ..... Date : .....

Signature : .....

Amitavo Chowdhury Wye
Authorized Person
ShipDyn Ltd.

ROUNAK SAHA NILOY
Naval Architect and Marine Engineer
Designer
ShipDyn Ltd.

Plan Approval Procedure (Inland Vessels)

Following procedures should be followed for approving plans for all types of inland vessels. The procedure ensure compliance with all the mandatory requirements specified by various inland vessels rules made under the ISO'76.

General Instructions:

1. All plans submitted must show sufficient details to demonstrate compliance with the relevant rules.

2. Structural Fire Safety-S.R.O. No. 301/Law, dt: 31.10.01, rues-22-35 regarding heat resistant lining, Exhaust piping, Sight glass, Pump room, Steer cage, Air intake/Exhaust ducting is to be followed and described clearly in the submission of drawing for approval.

3. Six sets of plan is to be submitted, including one soft copy in compact disc (CD), where feasible.

4. A list of documents and plans & an index shall be included with each submission;

5. 1:100 Scale to be used in all design & plan, except smaller plans which can be drawn using scale 1:50. In one plan no more than one scale can be used, all dimensions must be in mm.

6. Every plan is to be signed and stamped by the designer, person responsible to check the design, CEO of the design firm and the owner, with date.

7. All designs & plans and documents to be submitted with sequential number,

8. Section modulus calculation for applicable structures such as kilson, floor plate scantling, kilsons & webs, shell & deck scantling, side frame, etc must be submitted along with the drawings.

9. Stability calculation as per Stability rules for inland ship (S.R.O. No. 287/Law, dt:30.10.02) is to be submitted as follows: The underwater shape and KM calculation done- -either manually, or by using computer software. If manually, details of the calculation to be shown separately. If using computer software- software type approval certificate to be submitted to the DOS, that has been approved by any IACS (International Associate of Classification Societies) member society. Requirements of calculation: - Ships up to 70 metre, offset table interval max. 1 cm, based upon the offset table, under water shape and KM calculation to be prepared. - Calculation sheet should calculation of under water volume, water plan area at different draught, calculation of all hydrostatic curve, section modulus. Note: The Stability criteria for inland vessel need to be established or yet to be established, Till that time vessel's actual stability criteria, calculation manually or by using computer software, to be shown in the proposed plan and need not to be complied with the criteria set in the rule book (IMO criteria)

10. Demonstration or Compliance with the crowing requirement: With all passengers (full capacity & 75 kg/passenger) on one extreme side on top deck vessel's heeling must not exceed 10 degree.

11. General Characteristics for main and auxiliary engine and equipments to be submitted and Placing and ventilation of accumulation to be shown.

Amitavo Chowdhury Wye
Authorized Person
ShipDyn Ltd.

ROUNAK SAHA NILOY
Naval Architect and Marine Engineer
Designer
ShipDyn Ltd.

12. Plans to be drawn by any design firm/ competent naval architect having a qualification approved by BUET or having a membership of any international professional body of Naval Architecture. The designing firm/person shall submit the plan for approval with a bank guarantee of Tk. 5, 00,000/- as security on behalf of the firm/person.

General requirements:

1. Designs & Plans for approval of Inland Vessel to be submitted to the Department of Shipping for inland vessels of Length not exceeding 70 meter,

2. Passenger Launch can be maximum tow decks only (main deck & second deck),except bridge deck of vessel exceeding 50 meter length,

3. Bridge deck of passenger launch (above 50 meter length) will contain only navigating bridge and no accommodation, situated not exceeding and with in (1/3) one third length of vessel, length from forward.

4. All topmost deck including second deck and bridge will be of half rounded design,

5. All opening in the hull and main deck including Engine Room to be of weather tight design & feathers of the water tightness (closings) to be shown in the plan.

6. Plan should include main machinery specification including main propulsion machinery electrical generator, emergency source of power, type of main and emergency steering arrangement (if any), number(s) of proposed main engines/aux, engine, number of cylinders, cylinder diameter, kilo watt power, make, power versus speed curve, maximum speed of vessel at designed power considering water & wind resistances,

Design governing Criteria:

LOA=	76.21 M
B=	15.24 M
D=	4.87 M
Frame spacing=	550 mm
Type=	Steel Body Cargo Vessel
Material of construction=	Steel (Old)
Area of operation.=	Inland & bay crossing

Amitavo Chowdhury Wye
Authorized Person
ShipDyn Ltd.

ROUNAK SAHA NILOY
Naval Architect and Marine Engineer
Designer
ShipDyn Ltd.

Verification for designs & Plans submitted to the Department of Shipping as follows:

	For official use only Complied with		
	Yes	No	N/A
01. Length of Vessel- within 70 meters — Yes	☐	☐	☐
02. Number of decks-..... 3	☐	☐	☐
03. Number of decks of passenger launch-- ------N/A	☐	☐	☐
04. Bridge deck on top of second deck-- Yes	☐	☐	☐
05. Ship's deck length > 50m	☐	☐	☐
1. Bridge deck contain only navigation bridge-----Yes	☐	☐	☐
2. Any accommodation on bridge deck----------No	☐	☐	☐
3. Bridge deck within 2/3 of vessel length------	☐	☐	☐
06 . All topmost decks are half rounded design—N/A	☐	☐	☐
07. All openings are weather tight design & features of water	☐	☐	☐
tightness (closing) shown in the plan------Yes	☐	☐	☐
08. Plans are numbered sequentially---------- Yes	☐	☐	☐
09. Scale used-as per requirement----------Yes	☐	☐	☐
10. Machinery specifications included----------No	☐	☐	☐
11. Electrical Power (Main & Emergency)	☐	☐	☐
specifications included - No	☐	☐	☐
12. Steering power	☐	☐	☐
13. Steering Mechanical	☐	☐	☐
14. Plan is signed and stamped by -	☐	☐	☐
01. Owner : MOHAMMAD HOSAIN SUMON	☐	☐	☐
S/O ABDUS SOBAN KHALASHI	☐	☐	☐
02. Drawn :	☐	☐	☐
03.Designer : ROUNAK SAHA NILOY	☐	☐	☐
04.Checker : AMITAVO CHOWDHURY WYE	☐	☐	☐
15. Submitted six sets of plan ---------- Yes	☐	☐	☐
16. Submitted one CD version soft copy of plan ---------- Yes	☐	☐	☐
17. Contents & index submitted ----------Yes	☐	☐	☐

Amitavo Chowdhury Wye
Authorized Person
ShipDyn Ltd.

ROUNAK SAHA NILOY
Naval Architect and Marine Engineer
Designer
ShipDyn Ltd.

1. Vessel Particulars :

(a) Vessel Name	:	M. V. YOUNUS MOLLA-1
(b) L OA	:	76.21 M
(c) L BP-	:	69.85 M
(d) L WL-	:	74.29 M
(e) B EXT-	:	15.26 M
(f) B MLD-	:	15.24 M
(g) DEPTH EXT-	:	4.89 M
(h) DEPTH MLD-	:	4.87 M
(i) DRAUGHT-	:	4 M
(j) Number of propeller	:	2
(k) Diameter of propeller	:	1505 MM
(l) Type of Engines	:	MARINE DIESEL ENGINE, CHINA
(m) Power of Engines	:	520 X 2 H.P.

2. Type of Vessel –

Passenger ☐ Cargo ☑ Tanker ☐ Barge ☐ Ferry ☐ Others ☐

3. Owner -	:	MOHAMMAD HOSAIN SUMON
Company & Address	:	M/S. SOHA MARINE RESOURCES
Designer -	:	ROUNAK SAHA NILOY
Checked by -	:	AMITAVO CHOWDHURY WYE
Designing firm name	:	Shipdyn Ltd.
Designing firm address	:	128, Malek Mansion, Motijheel C/A, Dhaka-1000

Material - :

Steel (New) ☐ Steel (Old) ☑ Wood ☐ Mixed ☐ Glass fiber ☐ Aluminum ☐

4. Proposed route :

Inland (calm water)	☐	Inland (partially smooth water)	☐
Inland (rough water)	☐	Inland (round seasonal water)	☐
Inland & bay crossing	☑	Inland & seasonal bay crossing	☐

5. Plying time :

Day ☐ Night ☐ Day & Night ☑

Amitavo Chowdhury Wye
Authorized Person
ShipDyn Ltd.

ROUNAK SAHA NILOY
Naval Architect and Marine Engineer
Designer
ShipDyn Ltd.

Verification of drawing & documents submitted for approval - For official use only

Details of the drawing	Comments (Yes / No)
1.General Arrangement : Bulwark, railing, accommodation layout, deck layout, Engine room layout	☐ ☐
2. Safety Plan:To include Fire Extinguishing System, the position of fire hydrant,nozzle,	☐ ☐
fire pump, portable fire extinguishers, fire bucket, axe, sand box, crow bar, CO2 room,	☐ ☐
deck isolating valve, shore connection,breathing apparatus,etc.and the easiest passage	☐ ☐
route to reach the fire fighting appliances in case of emergency. (Fire & Safety Plan	☐ ☐
S.R.O. No.301/Law, dt: 31.10.01,rule-1-35 is to be followed and described clearly in the	☐ ☐
submission of drawing for approval.	☐ ☐
All bulkheads in accommodation is to be of min. 5 mm (new plate and 20% extra for	☐ ☐
old plates ;)	☐ ☐
Life Saving Appliances (LSA) Safety Plan: (S.R.O No. 306/Law, dt:31.10.01) All the life	☐ ☐
saving appliance & equipment and easiest way to reach to the L.S.A shall be properly	☐ ☐
marked and shown in the plan & design for approval.	☐ ☐
3. Lines Plan: To include off-set table, etc.	☐ ☐
4. Hydrostatic curves: To include hydrostatic curve & table of dead weight, cf, cb, cm,	☐ ☐
cp, km, kn, etc.	☐ ☐
5. Cross Curve of stability: At interval of at least 15 degree.	☐ ☐
6. Preliminary stability booklet: Stability booklet shall contain at least following information:	☐ ☐
• A table of contents & index for the booklet,	☐ ☐
• General description of the ship with principal dimensions	☐ ☐
• General arrangement plan showing water tight compartment, permanent ballast if any, freeboard diagram.	☐ ☐
• Hydrostatic curves/ tables & cross curve of stability calculated on even keel and intact & dynamic stability.	☐ ☐
• Capacity plan/ table showing capacity and CG of each compartment	☐ ☐
• Tank sounding table showing capacity and CG of each tank.	☐ ☐
• Information on loading restriction, such as , maximum KG or Maximum GM curve or table that can be used to determine compliance with the applicable stability criteria.	☐ ☐
• At least four standard operating condition & examples (as per Inland Ship Stability rule-2001) for developing other loading condition using information contained in the stability booklet.	☐ ☐

Amitavo Chowdhury Wye
Authorized Person
ShipDyn Ltd.

ROUNAK SAHA NILOY
Naval Architect and Marine Engineer
Designer
ShipDyn Ltd.

Verification of drawing & documents submitted for approval - For official use only

Details of the drawing	Comments (Yes / No)
• Details calculation showing whether the vessel satisfy the specific requirement of " Inland Shipping Rule (Stability) 2001"	☐ ☐
• Guidance for the master for safe operation of the ship.	☐ ☐
• Any other information which are unique/peculiar to the vessel.	☐ ☐
• Simplified stability date for passenger ships. Stability calculation to be shown for the following conditions:	☐ ☐
a) All passenger on one side with no wind and maximum beam wind speed of 60 knot & gust of 100 knots.	☐ ☐
• (b) Half passenger on onbe side with no wind and maximum beam wind speed of 60 knots & gust of 100 knots	☐ ☐
• (c) Empty ship with no wind and maximum beam wind speed of 60 knots & gust of 100 knots,	☐ ☐
In all cases vessel must have enough reserve bouncy to be upright.	☐ ☐
7. Capacity plan: Capacity plan must show the k.g of each compartment, floor area of the compartment, cubic capacity of the compartment.	☐ ☐
8. Mid ship section: ½ midship seetion drawing must show bilge curvature & extension of at least three cross section, kilsons, web frame, stringer, etc	☐ ☐
9. Drawing of longitudinal sections and deck plans including seating and openings: To include manholes, scupper, freeing port, kilsons, web frame, stringer, etc.	☐ ☐
10. Shell expansion including welds, openings, stringers, frames etc: To include plate thickness, frame thickness, frame spacing. Etc	☐ ☐
11. Hatch coaming and covers: Drawing to include hatch way beam, hatch cover, deck plating, coaming height, hatch dimension, hatch cover dimension, insert plate.	☐ ☐
12. Bulkheads including tank bulkheads- To include dimensions and spacing of stiffeners and plate thickness.	☐ ☐
13. Tank top and separate tanks: To show manholes, soundings pipes, vent pipes, fillings pipes etc.	☐ ☐
14. Bottom structure in engine room- To show engines foundations.	☐ ☐
15. Main engine, Auxiliary machinery and equipment, deck equipment, davit etc. foundations	☐ ☐
16. Rudder & Steering arrangement: To show details of the rudder plate, rudder post, rudder stock, rudder stopper, main and emergency steering, coupling flange, pintle, propeller post, propeller boss and the dimensions, tiller and quadrant dia, steering chain, rod, sole piece, heel piece etc.	☐ ☐
17. Calculation of equipment number:	☐ ☐
18. Anchoring and mooring arrangement: To show anchor chain dimensions, length, chain dia , breaking load.	☐ ☐

Amitavo Chowdhury Wye
Authorized Person
ShipDyn Ltd.

ROUNAK SAHA NILOY
Naval Architect and Marine Engineer
Designer
ShipDyn Ltd.

Verification of drawing & documents submitted for approval - For official use only

Details of the drawing	Comments (Yes / No)
19. Rigging : To show the position of mast, mast head light, side light, stern lights, anchor light, towing light, not under command (NUC) lights, dangers cargo operation light, signal light with height above the main deck, vertical & horizontal separation and position & numbers of search light and whistles, intercom position, bridge view and window positioning for clear aft view. (S.R.O No. 305/ Law dt: 30.10.01 (Navigations) * Dangers cargo operation light, Intercom Position- Not Provided.	☐ ☐
20. Super Structure and deck house: To include stores, common spaces, galley, No. of dead light, port holes, sill height, air, filling and vent pipe height etc	☐ ☐
21. General arrangements of machinery spaces.	☐ ☐
22. Shafting and propeller: To show propeller boss, propeller details, shaft details.	☐ ☐
23. Sounding, piping diagrams for bilge, emergency bilge injection, fire fighting, air, fuel, oil, lubricating oil, deck washing and all pipes under pressure: To show the position of hydrants, over flow pipes, air pipes, soundings pipe, pipe dimensions, striking plates, bilge and ballast piping.	☐ ☐
24.Cargo piping and pump room including ventilation for tankers	☐ ☐
25.Electrical installations: To show wiring for steering gear and navigation lights, cabins, engine room, position of switch board, etc	☐ ☐
26. Accommodation plan Officers, crew & passengers: The drawing mast show sky light, angle, elevation, number, position & dimensions of door and stairs, Positions and numbers of W.C tiolets, basins & bath rooms, dressing rooms for ladies & gents, The railing with number of rails and uprights, spacing of railings and uprights (max. spacing for rails-200 mm, uprights-900 mm), Hands railings for all stairs more than three steps,(Emergency exit minimum15 mm/passenger) Including of stairs: For passenger-max. 40 degree For crew members-max 50 degree Engine room, pump room & other working spaces-max 55 degree	☐ ☐
27.Chain locker & Hatch in the way of tank	☐ ☐
28. Aft & Fore body construction	☐ ☐
29. Details of Propeller	☐ ☐

Amitavo Chowdhury Wye
Authorized Person
ShipDyn Ltd.

ROUNAK SAHA NILOY
Naval Architect and Marine Engineer
Designer
ShipDyn Ltd.

			For office use only Complied with Yes	No
7. Hull Plate material dimensions (rule-8)				
min. -4. (new plate)	As per rule ....…	**5.60 mm**	☐	☐
min.-6 (old plate)	As per drawing no. ...08... ..	**6 mm**	☐	☐
8. Keel Plating (rule 9)				
Requirement as per table-1 of the	As per rule	**11.19 mm**	☐	☐
schedule of sro. no. -308/law-2001	As per drawing no. 08	**12 mm**	☐	☐
9. Bottom Plating (rule-10)				
Requirement as per table-2 of the	As per rule	**10.49 mm**	☐	☐
schedule of sro. no. -308/law-2001	As per drawing no. 08	**12 mm**	☐	☐
10. Frame Spacing (standard-500mm) (min.-450mm, max.-550mm) (rule-25)				
Section modulus requirement as	As per rule	**279.16 cm³**	☐	☐
per table-17 of the schedule	As per cal. sheet no. 1/cal	**349.69 cm³**	☐	☐
11. Bilge plating (rule 11)				
Requirement as per table-3	As per rule	**13.29 mm**	☐	☐
of the schedule	As per drawing no. 08	**14 mm**	☐	☐
12. Side shell plating (rule-12)				
Requirement as per table-4	As per rule	**10.49 mm**	☐	☐
of the schedule	As per drawing no. 08	**12 mm**	☐	☐
13. Deck plating (Hatch Way plate) (rule-13)				
Requirement as per table-5	As per rule	**12.59 mm**	☐	☐
of the schedule	As per drawing no. ….....08	**13 mm**	☐	☐
14. Deck plating (All deck plate) (rule-13)				
Requirement as per table - 6	As per rule	**8.39 mm**	☐	☐
of the schedule	As per drawing no. ….....08	**9 mm**	☐	☐
15. Deck plating (Double Bottom inner plate) (rule-13)				
Requirement as per table - 7	As per rule	**N/A**	☐	☐
of the schedule	As per drawing no. ….....	**N/A**	☐	☐

Amitavo Chowdhury Wye
Authorized Person
ShipDyn Ltd.

ROUNAK SAHA NILOY
Naval Architect and Marine Engineer
Designer
ShipDyn Ltd.

			For office use only Complied with Yes	No
16. Stem (rule-14)				
Requirement as per table - 7	As per rule	**15.38 mm**	☐	☐
of the schedule	As per drawing no. ……10	**16 mm**	☐	☐
17. Stem Bar (rule-15)				
Requirement as per table - 8	As per rule	**164.80 mm**	☐	☐
of the schedule	As per drawing no.....	**164.80 mm**	☐	☐
18. Stern frame breadth (rule-16)				
Requirement as per table - 9	As per rule	**164.80 mm**	☐	☐
of the schedule	As per drawing no.....	**170 mm**	☐	☐
19. Stern frame thickness (rule-16)				
Requirement as per table - 9	As per rule	**44.92 mm**	☐	☐
of the schedule	As per drawing no.	**40 mm**	☐	☐
20. Propeller Post breadth (rule-17)				
Requirement as per table-10	As per rule……..	**274.60 mm**	☐	☐
of the schedule	As per drawing no.	**190 mm**	☐	☐
21. Propeller Post radius(rule-17)				
Requirement as per table-10	As per rule……..	**44.92 mm**	☐	☐
of the schedule	As per drawing no.	**75 mm**	☐	☐
22. Propeller Post thickness (rule 17)				
Requirement as per table-10	As per rule……..	**74.83 mm**	☐	☐
of the schedule	As per drawing no.	**52 mm**	☐	☐
23. Rudder post (rule-18)				
Requirement, welded, (min 4 boll), rudder	As per rule……..	**4 bolts**	☐	☐
can be sold or streamlined plate rudder (hollow)	As per drawing no ..16	**6 bolts**	☐	☐
24. Sole Piece breadth (rule-19)				
Requirement as per table-11	As per rule	**209.70 mm**	☐	☐
of the schedule	As per drawing no……....16	**209.70 mm**	☐	☐

Amitavo Chowdhury Wye
Authorized Person
ShipDyn Ltd.

ROUNAK SAHA NILOY
Naval Architect and Marine Engineer
Designer
ShipDyn Ltd.

			For office use only Complied with Yes	No
25. Sole Piece thickness (rule-19)				
Requirement as per table-11	As per rule	**74.90 mm**	☐	☐
of the schedule	As per drawing no. ..... 16.	**74.90 mm**	☐	☐
26. Heel piece (rule-20)				
Requirement :Heel piece length shell be	As per rule	**1650 mm**	☐	☐
minimum three times the frame spacing	As per drawing no….	**17828 mm**	☐	☐
27. Propeller boss (rule-21)				
Requirement as per table-12	As per rule	**38 mm**	☐	☐
of the schedule	As per drawing no……..22	**38 mm**	☐	☐
28. Bottom construction (Floor plate scantling/section modulus) (rule-22)				
Requirement as per table-13	As per rule	**2838.56 cm³**	☐	☐
of the schedule	As per cal sheet no. ……2/cal	**1239.57 cm³**	☐	☐
	As per drawing no. 08	**1239.57 cm³**	☐	☐
29. Kilsons (rule-23)				
Requirement as per table-14	As per rule	**5 ……Nos**	☐	☐
of the schedule	As per drawing no…….....08	**5 ……Nos**	☐	☐
30. Webs thickness & sectional area (rule-23)				
Requirement as per table-15	As per rule	**8.99 mm**	☐	☐
of the schedule	As per drawing no……08	**11 mm**	☐	☐
31. Kilsons & Webs (Centre & side girders thickness & sectional area) (rule-23)				
Requirement as per table-16	As per rule	**9.49 mm**	☐	☐
of the schedule	As per drawing no……08	**12 mm**	☐	☐
32. Shell & deck scantling (section modulus) (rule-24)				
Requirement as per table-13	As per rule…….....	**279.16 cm³**	☐	☐
of the schedule	As per cal sheet no. ……2/cal	**349.69 cm³**	☐	☐
33. Side frame (sectional modulus) (rule-25)				
Requirement as per table-17	As per rule…….....	**279.16 cm³**	☐	☐
of the schedule	As per cal. sheet. no. 01/cal	**349.69 cm³**	☐	☐

Amitavo Chowdhury Wye
Authorized Person
ShipDyn Ltd.

ROUNAK SAHA NILOY
Naval Architect and Marine Engineer
Designer
ShipDyn Ltd.

			For office use only Complied with Yes	No
34. Frame spacing : (standard-500mm) (min.-450mm, max.-550mm) (rule-25)				
Drawing must include specification	As per rule…….....	**589.70 mm**	☐	☐
of frame to ensure section modulus	As per cal. sheet no.03/cal.	**550 mm**	☐	☐
35. Web frame (Transverse framing difference between web framing) (rule-26)				
Requirement as per table-18 of the	As per rule…….....	**297.88 cm³**	☐	☐
schedule of sro.no-308/law-2001	As per drawing no…....01	**471.50 cm³**	☐	☐
36. Side stringer (between web framing (if d>2.5m) (rule-27)				
Requirement as per table-19 of the	As per rule…….....	**297.88 cm³**	☐	☐
schedule of sro.no-308/law-2001	As per cal. sheet no. 04/cal	**432.49 cm³**	☐	☐
37. Deck beam (section modulus) (rule-28)				
Requirement as per table-20 of the	As per rule…….....	**18.67 cm³**	☐	☐
schedule of sro.no-308/law-2001	As per cal. sheet no.....05/cal	**128.14 cm³**	☐	☐
38. Deck web beam (section modulus) (rule-29)				
Requirement must be twice of the	As per rule…….....	**37.34 cm³**	☐	☐
deck beam as per table-20 of the	As per cal. sheet. no. 06/cal	**813.65 cm³**	☐	☐
39. Deck girder (section modulus) (rule-30)				
Requirement as per table-21 of the	As per rule…….....	**877.60 cm³**	☐	☐
schedule of sro.no-308/law-2001	As per cal. sheet. no. 07/cal	**813.65 cm³**	☐	☐
40. Pillar (sectional modulus) (rule-31)				
Requirement as per table-22 & 23 of the	As per rule…….....	**101.60 mm**	☐	☐
schedule of sro.no-308/law-2001	As per cal. sheet no.	**101.60 mm**	☐	☐
41. Bulkhead (rule-32)				
if l> 20m, minimum number of bulkhead is 2	As per rule	**7.70 mm**	☐	☐
if l> 30m, 2 bulkhead fore & aft. engine room,1	Asper drawing no.01	**10 mm**	☐	☐

Amitavo Chowdhury Wye
Authorized Person
ShipDyn Ltd.

ROUNAK SAHA NILOY
Naval Architect and Marine Engineer
Designer
ShipDyn Ltd.

			For office use only Complied with Yes	No
42. Collision Bulkhead (position) : (rule-33)				
Requirement as per table-24	As per rule……	**3.49 m**	☐	☐
schedule of sro.no-308/law-2001	As per cal. sheet. no.08/cal	**4.80 m**	☐	☐
43. After peak Bulkhead (section modulus) (rule-33) :				
Requirement as per table-25 of the	As per rule……	**357.70 cm³**	☐	☐
schedule of sro.no-308/law-2001	As per cal. sheet no. ...09/cal	**370.93 cm³**	☐	☐
44. Other Bulkhead (section modulus) (rule-34)				
Requirement as per table-26 of the	As per rule……	**357.70 cm³**	☐	☐
schedule of sro.no-308/law-2001	As per cal. sheet no. ..10/cal	**370.93 cm³**	☐	☐
45. Other Bulkhead thickness & sectional modulus : (rule-34)				
Requirement as per table-26 of the	As per rule……	**7.70 mm**	☐	☐
schedule of sro.no-308/law-2001	As per drawing no. 12	**10 mm**	☐	☐
Requirement as per table-26 of the	As per rule……	**7.70 mm**	☐	☐
schedule of sro.no-308/law-2001	As per cal. sheet no…..10/cal	**10 mm**	☐	☐
46. Bulkhead penetration: (rule-35)				
Collision bulkhead must not have any	As per rule……	**7.70 mm**	☐	☐
door or penetration, other bulkhead	As per drawing no..	**10 mm**	☐	☐
47. Water tight door (rule-36)				
Number of water tight door is to keep	As per rule..As keep as possible..Min.no.of blkhd pent	**7.70 mm**	☐	☐
48. Hull Tanks: (rule-37)				
Requirement as per table-27	As per rule……	**7.70 mm**	☐	☐
of the schedule	As per drawing no…..	**10 mm**	☐	☐
Requirement as per table-27		**450… …..mm**	☐	☐
of the schedule	As per rule	**7.70 mm**	☐	☐

Amitavo Chowdhury Wye
Authorized Person
ShipDyn Ltd.

ROUNAK SAHA NILOY
Naval Architect and Marine Engineer
Designer
ShipDyn Ltd.

			For office use only Complied with Yes	No
49. Separate tanks: (rule-38)				
Plate thickness	As per rule ....4.....mm	**7.70 mm**	☐	☐
Note: Separate tank must be fitted with framing and secured properly. No	As per drawing no..... ....6.....mm	**10 mm**	☐	☐
50. Drinking water tank (rule-39)				
Tank separated by	As per rule……	**7.70 mm**	☐	☐
Fill, vent, sounding pipe	As per drawing no. 13……	**10 mm**	☐	☐
separate	As per rule……	**7.70 mm**	☐	☐
Note: Drinking water tank must be separated from other tank by cofferdam. No sanitary	As per drawing no…	**10 mm**	☐	☐
51. Wash Bulkhead (rule - 40)				
plate thickness-min. 5 mm	As per rule…….	**7.70 mm**	☐	☐
section modulus min. 5 cu. Cm	As per drawing no.	**10 mm**	☐	☐
Note: Stiffener to run till second frame	As per rule…….	**7.70 mm**	☐	☐
	As per cal. sheet no	**10 mm**	☐	☐
52. Engine room keelsons (sectional area) (rule-44)				
Requirement as per table-15	As per rule…….	**16.80 mm**	☐	☐
of the schedule	As per cal sheet. no. 11/cal..	**18 mm**	☐	☐

Amitavo Chowdhury Wye
Authorized Person
ShipDyn Ltd.

ROUNAK SAHA NILOY
Naval Architect and Marine Engineer
Designer
ShipDyn Ltd.

			For office use only Complied with Yes	No
53. Engine room web frames (web distance) (rule-45)				
Requirement as per table-18	As per rule…….	**297.88 cm³**	☐	☐
of the schedule	As per drawing no. 14…..	**471.50 cm³**	☐	☐
54. Engine room web frames (section modulus) (rule-45)				
Requirement as per table-29	As per rule…….	**297.88 cm³**	☐	☐
of the schedule	As per cal. sheet no. 12/cal	**471.50 cm³**	☐	☐
55. Longitudinal hatch coming (rule-47)				
Requirement as per table-30	As per rule…….	**11.89 mm**	☐	☐
of the schedule	As per drawing. no. ..11	**13 mm**	☐	☐
56. Transverse hatch coming (rule-48)				
Requirement as per table-31	As per rule…….	**11.89 mm**	☐	☐
of the schedule	As per drawing. no. ..11	**13 mm**	☐	☐
57. Hatch cover auxiliaries (rule-49)				
Hatch cover breadth	As per rule…….	**18.20 cm³**	☐	☐
	As per drawing. no. ..11.	**72.53 cm³**	☐	☐
58. Hatch cover auxiliaries (section modulus) (rule-49)				
Requirement as per table-32	As per rule…….	**18.20 cm³**	☐	☐
of the schedule	As per cal. sheet. no. ..13/cal	**72.53 cm³**	☐	☐
59. Hatch way beam (rule-50)				
Hatch cover beam	As per rule…….	**18.20 cm³**	☐	☐
	As per cal. sheet. no. ..14/ca	**72.53 cm³**	☐	☐
60. Hatch way beam (section modulus) : (rule-50)				
Requirement as per table-33	As per rule…….	**18.20 cm³**	☐	☐
of the schedule	As per cal. sheet. no. ..14/ca	**72.53 cm³**	☐	☐
61. Side hatch way beam (rule-51)				
Side hatch way beam	As per rule…….	**18.20 cm³**	☐	☐
	As per cal. sheet. no. ...15/cal	**72.53 cm³**	☐	☐
62. Side hatch way beam (section modulus) : (rule-51)				
Requirement as per table-33	As per rule…….	**18.20 cm³**	☐	☐
of the schedule	As per cal. sheet. no. ...15/ca	**72.53 cm³**	☐	☐

Amitavo Chowdhury Wye
Authorized Person
ShipDyn Ltd.

ROUNAK SAHA NILOY
Naval Architect and Marine Engineer
Designer
ShipDyn Ltd.

			For office use only Complied with Yes	No
63. Steel Hatch cover (rule-52)				
Min cover thickness-4 mm	As per rule…….	**6 mm**	☐	☐
20% extra for bay crossing	As per drawing. no. ..11	**8 mm**	☐	☐
64. Steel hatch cover (section modulus) : (rule-52)				
Requirement as per table-34	As per rule…….	**7 cm³**	☐	☐
of the schedule	As per cal. sheet. no.16/cal	**55.46 cm³**	☐	☐
65. Wooden hatch cover : (rule-53)				
Wooden hatch cover	As per rule…….	**N/A**	☐	☐
Length	As per drawing. no.	**N/A**	☐	☐
Wooden hatch cover	As per rule…….	**N/A**	☐	☐
thickness	As per drawing. no.	**N/A**	☐	☐
Securing arrangement	As per rule…….	**....Batten..... mm**	☐	☐
by		**.....Batten..... mm**	☐	☐
66. Deck opening : (rule-54)				
Deck opening corners to be	As per rule…….	**11.89 mm**	☐	☐
properly strengthened	As per drawing no. ..11	**13 mm**	☐	☐
with	As per rule…….	**11.89 mm**	☐	☐
		complied mm	☐	☐
67. Doors on main deck (rule-55)				
Door sills	As per rule…….	**7.70 mm**	☐	☐
watertight Securing	As per drawing no. ...20	**10 mm**	☐	☐
arrangement with	As per rule…….	**7.70 mm**	☐	☐
		be provided mm	☐	☐
		done mm	☐	☐
68. Machinery space opening (rule-56)				
water tight casing height	As per rule…….	**7 mm**	☐	☐
space separated by	As per drawing no. ..20…	**7 mm**	☐	☐
space separated by	As per rule…….	**7 mm**	☐	☐
	As per rule…….	**7 mm**	☐	☐

Amitavo Chowdhury Wye
Authorized Person
ShipDyn Ltd.

ROUNAK SAHA NILOY
Naval Architect and Marine Engineer
Designer
ShipDyn Ltd.

			For office use only Complied with Yes	No
69. Opening at cargo hold (rule-57)				
Entry at cargo hatch	As per rule…….	**11.89 mm**	☐	☐
Toggles with water tight	As per drawing no. ...11	**13 mm**	☐	☐
steel hatch covers	As per rule…….	**11.89 mm**	☐	☐
	As per drawing no. 27	**13 mm**	☐	☐
70. Man hole & flush deck plug (rule-59)				
All man hole & flash deck plug to be water tight, if needed with gasket.	As per rule…….	**8.39 mm**	☐	☐
cover by	As per rule…….	**8.39 mm**	☐	☐
71. Freeing port (rule-60)				
Freeing port min. area	As per rule…….	**4.54 m²**	☐	☐
Nos. verticals stand	As per drawing no. 09	**10.53 m²**	☐	☐
are fitted in every	As per rule…….	**N/A**	☐	☐
	As per drawing no……	**N/A**	☐	☐
72. Drainage pipe (rule-61)				
Drainage pipe	As per rule…….	**Sufficient nos**	☐	☐
Drainage pipe thickness	As per drawing no. ..09	**Sufficient nos**	☐	☐
	As per rule…….	**3.05 mm**	☐	☐
	As per drawing no. 09	**10 mm**	☐	☐
73. Super structure, Deck house & quarter deck bulk head (rule-62)				
stiffened by	As per rule…….	**7.70 mm**	☐	☐
		complied. mm	☐	☐
74. Super structure, Deck house & Quarter deck bulkhead (rule-63):				
Bulkhead thickness	As per rule…….	**5.50 mm**	☐	☐
Bulkhead section modulus requirement	As per drawing no. .20.....	**10 mm**	☐	☐
as per table-35 of the schedule	As per rule…….	**16 cm³**	☐	☐
	As per cal. sheet. no. ..17/cal	**71.84 cm³**	☐	☐
75. Engine casing : (rule-64)				
Engine casing plate thickness	As per rule…….	**7 mm**	☐	☐
	As per drawing no. ...20	**7 mm**	☐	☐

Amitavo Chowdhury Wye
Authorized Person
ShipDyn Ltd.

ROUNAK SAHA NILOY
Naval Architect and Marine Engineer
Designer
ShipDyn Ltd.

			For office use only Complied with Yes	No
76. Decks (rule-65)				
Inside deck plate thickness	As per rule…….	**8.39 mm**	☐	☐
Outside deck plate thickness	As per drawing no. .. 20	**9 mm**	☐	☐
Deck beam section modulus requirement	As per rule…….	**8.39 mm**	☐	☐
as per table-36 of the schedule	As per drawing no. .. 20	**9 mm**	☐	☐
Deck girder section modulus requirement	As per rule…….	**18.67 cm³**	☐	☐
as per table-37 of the schedule	As per cal. sheet. no. .19/cal	**128.14 cm³**	☐	☐
	As per rule…….	**877.60 cm³**	☐	☐
	As per cal. sheet. no. ..20/cal	**813.65 cm³**	☐	☐
77. Rudder & steering gear (rule-66)				
Steering type	As per rule	**193.80 mm**	☐	☐
Emergency steering	As per drawing no. ...16	**193.80 mm**	☐	☐
	As per rule	**193.80 mm**	☐	☐
	As per drawing no. ...16	**193.80 mm**	☐	☐
78. Rudder stock (rule-67)				
Rudder area requirement as per	As per rule	**193.80 mm**	☐	☐
table 38& 39 of the schedule	As per drawing no. ...16	**193.80 mm**	☐	☐
Rudder stock dia requirement as per	As per rule	**193.80 mm**	☐	☐
table 38& 39 of the schedule	As per drawing no. ...16	**193.80 mm**	☐	☐
79. Pintle (rule-68)				
Pintle length	As per rule	**193.80 mm**	☐	☐
Pintle dia requirement	As per drawing no. ...16	**193.80 mm**	☐	☐
as per table-40 of the schedule	As per rule	**6.50 mm**	☐	☐
	As per drawing no. ...16	**10 mm**	☐	☐
80. Coupling flange (rule-69)				
Coupling flange thickness	As per rule	**30.20 mm**	☐	☐
	As per drawing no. ...16	**32 mm**	☐	☐

Amitavo Chowdhury Wye
Authorized Person
ShipDyn Ltd.

ROUNAK SAHA NILOY
Naval Architect and Marine Engineer
Designer
ShipDyn Ltd.

			For office use only Complied with Yes	No
81. Coupling bolt (rule-70)				
Rudder stock dia.	As per rule	**110 mm**	☐	☐
Coupling bolt dia.	As per drawing no. ...16	**125 mm**	☐	☐
	As per rule	**40.40 mm**	☐	☐
	As per drawing no. ...16	**32 mm**	☐	☐
82. Rudder plate (rule-71)				
plate rudder	As per rule	**10.49 mm**	☐	☐
	As per drawing no. ...16	**12 mm**	☐	☐
83. Main steering gear (rule-72)				
Rudder stock dia	As per rule	**193.80 mm**	☐	☐
Hard P to S	As per drawing no. ...16	**193.80 mm**	☐	☐
	As per rule	**193.80 mm**	☐	☐
	As per drawing no. ...16	**193.80 mm**	☐	☐
84. Emergency steering gear (rule-72)				
Type of steering gear	As per rule	**193.80 mm**	☐	☐
Hard P to S	As per drawing no….....	**193.80 mm**	☐	☐
3 position of steering	As per rule	**193.80 mm**	☐	☐
indicator	As per drawing no. …....	**193.80 mm**	☐	☐
	As per rule	**193.80 mm**	☐	☐
	As per drawing no…..	**193.80 mm**	☐	☐
85. Tillar & Quadrant : (rule-74)				
Hub dia requirement	As per rule	**199 mm**	☐	☐
as per rule-74(4)	As per drawing no. ...16	**220 mm**	☐	☐
bolt dia requirement as per	As per rule	**31 mm**	☐	☐
table-42 of the schedule	As per drawing no. ...16	**36 mm**	☐	☐
Tiller or quadrant dia. Requirement	As per rule	**1304 mm**	☐	☐
as per table-43 of the schedule	As per drawing no..16	**1110 mm**	☐	☐

Amitavo Chowdhury Wye
Authorized Person
ShipDyn Ltd.

ROUNAK SAHA NILOY
Naval Architect and Marine Engineer
Designer
ShipDyn Ltd.

			For office use only Complied with Yes	No
86. Steering chain & rod (rule-78)				
steering chain dia. Requirement	As per rule	**16 mm**	☐	☐
as per table-44 of the schedule	As per drawing no.....27.	**14 mm**	☐	☐
Steering rod dia requirement	As per rule	**18.20 mm**	☐	☐
as per table-44 of the schedule	As per drawing no.....27.	**17 mm**	☐	☐
87. Chain cable & ropes (accessories) (rule-79)				
Requirement	As per rule	**2 x 597 kg**	☐	☐
as per table-45	As per drawing no.	**2 x 1350 kg**	☐	☐
88. Requirement for chain cables (rule-81)				
Length of chain cable	As per rule	**275 m**	☐	☐
Chain dia requirement as per	As per drawing no. ...17	**275 m**	☐	☐
table-46 of the schedule	As per rule	**275 m**	☐	☐
bracking load requirement as per	As per drawing no. ...17	**275 m**	☐	☐
table-46 of the schedule	As per rule	**275 m**	☐	☐
	As per drawing no. ...17	**275 m**	☐	☐
89. Tow & mooring lines (rule-82)				
Mooring lines	As per rule	**275 m**	☐	☐
(Steel or synthetic)	As per drawing no. ...17	**275 m**	☐	☐
Length of tow/moor lines requirement	As per rule	**275 m**	☐	☐
as per table-46 of the schedule	As per drawing no. ...17	**275 m**	☐	☐
Dia. of lines requirement as per	As per rule	**275 m**	☐	☐
table-45 of the schedule	As per drawing no. ...17	**275 m**	☐	☐
90. Bulwark & railing (rule-83)				
Bulwark plate thickness requirement	As per rule	**5.99 mm**	☐	☐
as per table-47 of the schedule	As per drawing no. …09	**6 mm**	☐	☐

Amitavo Chowdhury Wye
Authorized Person
ShipDyn Ltd.

ROUNAK SAHA NILOY
Naval Architect and Marine Engineer
Designer
ShipDyn Ltd.

			For office use only Complied with Yes	No
91. Chain lockers (rule-84)				
Fore locker thickness requirement	As per rule	**6.98 mm**	☐	☐
as per table-48 of the schedule	As per drawing no. ...27	**8 mm**	☐	☐
Stern locker thickness	As per rule	**N/A**	☐	☐
	As per drawing no.	**N/A**	☐	☐
92. Propeller bracket (rule-85)				
V or I strut thickness requirement as per	As per rule	**38 mm**	☐	☐
table-49 or 50 of the schedule	As per drawing no.	**38 mm**	☐	☐
Shall plate thickness	As per rule	**N/A**	☐	☐
	As per drawing no	**N/A**	☐	☐
93. Anchor windlass (rule-86)				
Windlass (.....)	As per rule	**597 kg**	☐	☐
Nominal pull	As per drawing no. 17	**597 kg**	☐	☐
94. Mooring equipment (rule-87)				
Sets of double bollard	As per rule	**8 Nos**	☐	☐
Deck reinforced	As per drawing no. 01	**8 Nos**	☐	☐
	As per rule	**Doubler**	☐	☐
	As per drawing no. ..17	**Doubler**	☐	☐
95. Bottom sealing (rule-88)				
Wooden plank thickness	As per rule ……	**10.49 mm**	☐	☐
	As per drawing no. …..	**12 mm**	☐	☐
96. Bilge & ballast system : (rule-142)				
No. of bilge pump	As per rule ……	**2 nos**	☐	☐
	As per drawing no. ….21.	**2 nos**	☐	☐

Amitavo Chowdhury Wye
Authorized Person
ShipDyn Ltd.

ROUNAK SAHA NILOY
Naval Architect and Marine Engineer
Designer
ShipDyn Ltd.

			For office use only Complied with Yes	No
97. General service pump (rule-143)				
Gs p/p fitted	As per rule…….	**50 mm**	☐	☐
P/p drive	As per drawing no. 23	**50 mm**	☐	☐
	As per rule…….	**50 mm**	☐	☐
	As per drawing no……	**50 mm**	☐	☐
98. Air pipe : (rule-144)				
No. of tank or cofferdam	As per rule…….	**3.05 mm**	☐	☐
Position of air pipe	As per drawing no. ...23	**3.05 mm**	☐	☐
Closing fitted	As per rule…….	**3.05 mm**	☐	☐
No. of cargo hold	As per drawing no…..	**3.05 mm**	☐	☐
	As per rule…….	**3.05 mm**	☐	☐
	As per drawing no…..	**3.05 mm**	☐	☐
	As per rule…….	**3.05 mm**	☐	☐
	As per drawing no. 23	**3.05 mm**	☐	☐
99. Sounding pipe (rule-145)				
No of cargo hold	As per rule…….	**3.05 mm**	☐	☐
	As per drawing no. 23	**3.05 mm**	☐	☐

Amitavo Chowdhury Wye
Authorized Person
ShipDyn Ltd.

ROUNAK SAHA NILOY
Naval Architect and Marine Engineer
Designer
ShipDyn Ltd.

			For office use only Complied with Yes	No
100. Over flow pipe (rule-147)				
No. of over flow pipe	As per rule…….	**3.05 mm**	☐	☐
dia. of overflow pipe	As per drawing no. 23	**3.05 mm**	☐	☐
	As per rule…….	**3.05 mm**	☐	☐
	As per drawing no. 23	**3.05 mm**	☐	☐
101. Stability & KM calculation:				
Manually	Rule	**343.90 Z**	☐	☐
Using computer software	Drawing no.	**343.90 Z**	☐	☐
Software approved	Rule	**343.90 Z**	☐	☐
Approval. Cert. provided	Drawing no.	**343.90 Z**	☐	☐
102. Load lines & Free board calculation (rule 4-21)				
Load line & Free board	As per rule…….	**343.90 Z**	☐	☐
Sill height	As per drawing no. ..01	**343.90 Z**	☐	☐
Hatch coaming height	As per rule…….	**343.90 Z**	☐	☐
Air/ vent pipe height	As per drawing no. ….20	**343.90 Z**	☐	☐
	As per rule…….	**343.90 Z**	☐	☐
	As per drawing no. …..11	**343.90 Z**	☐	☐
	As per rule…….	**343.90 Z**	☐	☐
	As per drawing no. ……23	**343.90 Z**	☐	☐
103. Manhole/ entry to hatch-coaming height:				
Coaming height	As per rule…….	**600 mm**	☐	☐
	As per drawing no. ……27	**600 mm**	☐	☐
104. Passenger ship structural configuration (rule-214):				
Note: For ship above 40 meter longitudinal framing can be used at bottom, or as an	As per rule		☐	☐
and strength deck alternative transverse framing can be used at strength deck.	As per drawing no.		☐	☐

Amitavo Chowdhury Wye
Authorized Person
ShipDyn Ltd.

ROUNAK SAHA NILOY
Naval Architect and Marine Engineer
Designer
ShipDyn Ltd.

			For office use only Complied with Yes	No
Framing user at double				
bottom	As per drawing no. ……..	**N/A**	☐	☐
Framing used at deck				
	As per drawing no. ……..	**N/A**	☐	☐
Strength deck used				
	As per drawing no. ……..	**N/A**	☐	☐
D.B single compartment				
type	As per drawing no. ……..	**N/A**	☐	☐
105. Double bottom structure:				
Depth at centre line of	As per rule…….	**N/A**	☐	☐
D.B. (min. 800mm)	As per drawing no. ……..	**N/A**	☐	☐
Thickness of inner	As per rule…….	**N/A**	☐	☐
Plate (equal to bottom plate)	As per drawing no. ……	**N/A**	☐	☐
106. Double bottom bulkheads:				
Center line bulkhead (if	As per rule…….	**N/A**	☐	☐
B>8m)	As per drawing no. ……..	**N/A**	☐	☐
Transverse bulkhead in	As per rule…….	**N/A**	☐	☐
every D.B tank	As per drawing no. ……	**N/A**	☐	☐
Bulkhead plate thickness	As per rule…….	**N/A**	☐	☐
As described above the compliance of construction rules (S.R.O no. 308/Law,	As per drawing no. ……	**N/A**	☐	☐

Amitavo Chowdhury Wye
Authorized Person
ShipDyn Ltd.

ROUNAK SAHA NILOY
Naval Architect and Marine Engineer
Designer
ShipDyn Ltd.

Approval of design of an inland vessel

a. Vessel Name	:	M. V. YOUNUS MOLLA-1
b. L OA	:	76.21 M
c. L BP-	:	69.85 M
d. L WL-	:	74.29 M
e. B EXT-	:	15.26 M
f. B MLD-	:	15.24 M
g. DEPTH EXT-	:	4.89 M
h. DEPTH MLD-	:	4.87 M
I. DRAUGHT-	:	4 M
j. Number of propeller	:	2
k. Diameter of propeller	:	1505 MM
l. Type of Engines	:	MARINE DIESEL ENGINE, CHINA
m. Power of Engines	:	520 X 2 H.P.
n. Ship's owner name & address	:	MOHAMMAD HOSAIN SUMON (& others) M/S. SOHA MARINE RESOURCES
designed by (firm/person)	:	ROUNAK SAHA NILOY
checked by (firm/person)	:	AMITAVO CHOWDHURY WYE
verified & checked for	:	Compliance of the rules by (DOS/LCS/BIWTA)
approved by	:	Department of Shipping Dhaka.
approval no. & date	:	DOS/PLN/_____ Date: _____

The Department of Shipping hereby approves the plan of the above mentioned vessel on condition that the Construction Supervisor will ensure compliance of the Inland Shipping Ordinance 1976 & rules made there under and as per this approval booklet including instructions for supervisor during the construction of the vessel. He will also submit a compliances certificate after the completion and test trial of the vessel for registration of the vessel to the registrar of Inland Ships.

Amitavo Chowdhury Wye
Authorized Person
ShipDyn Ltd.

ROUNAK SAHA NILOY
Naval Architect and Marine Engineer
Designer
ShipDyn Ltd.